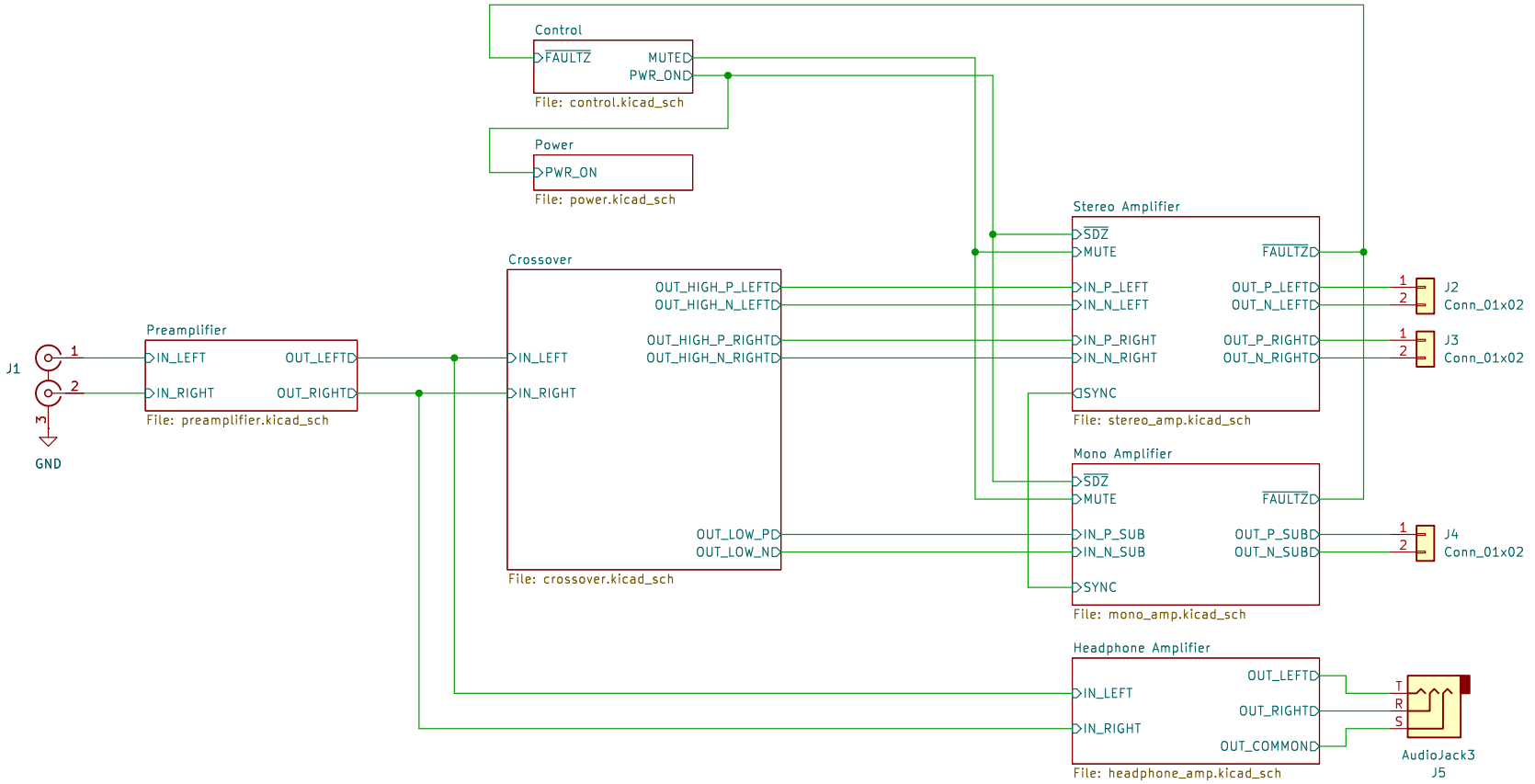
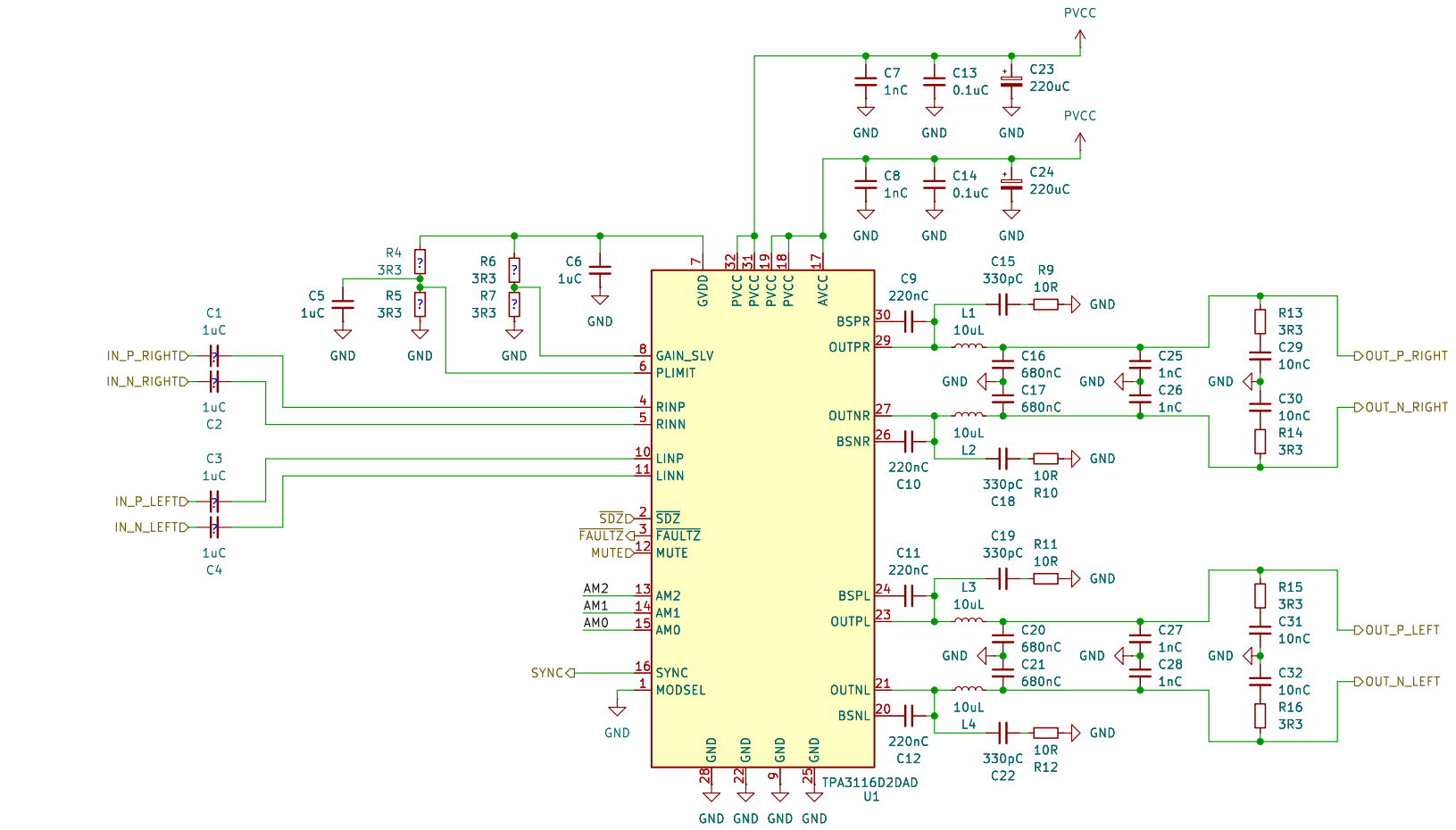
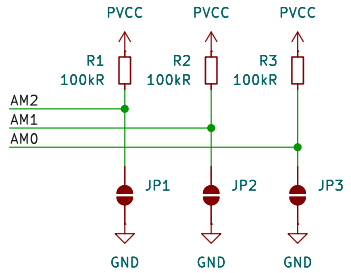
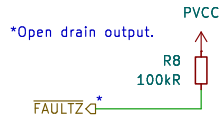


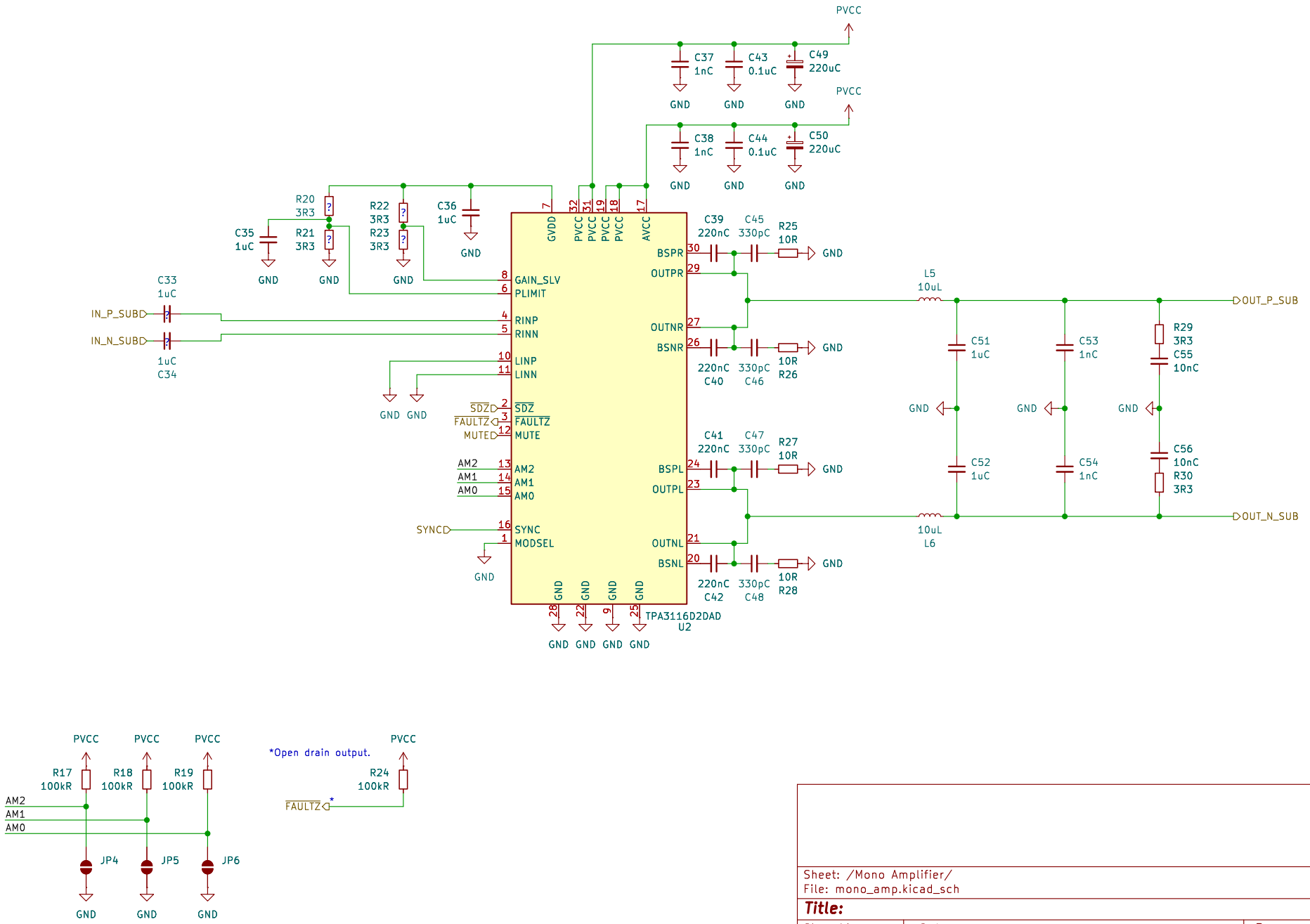
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Title:		
Size: A4	Date:	**Rev:**
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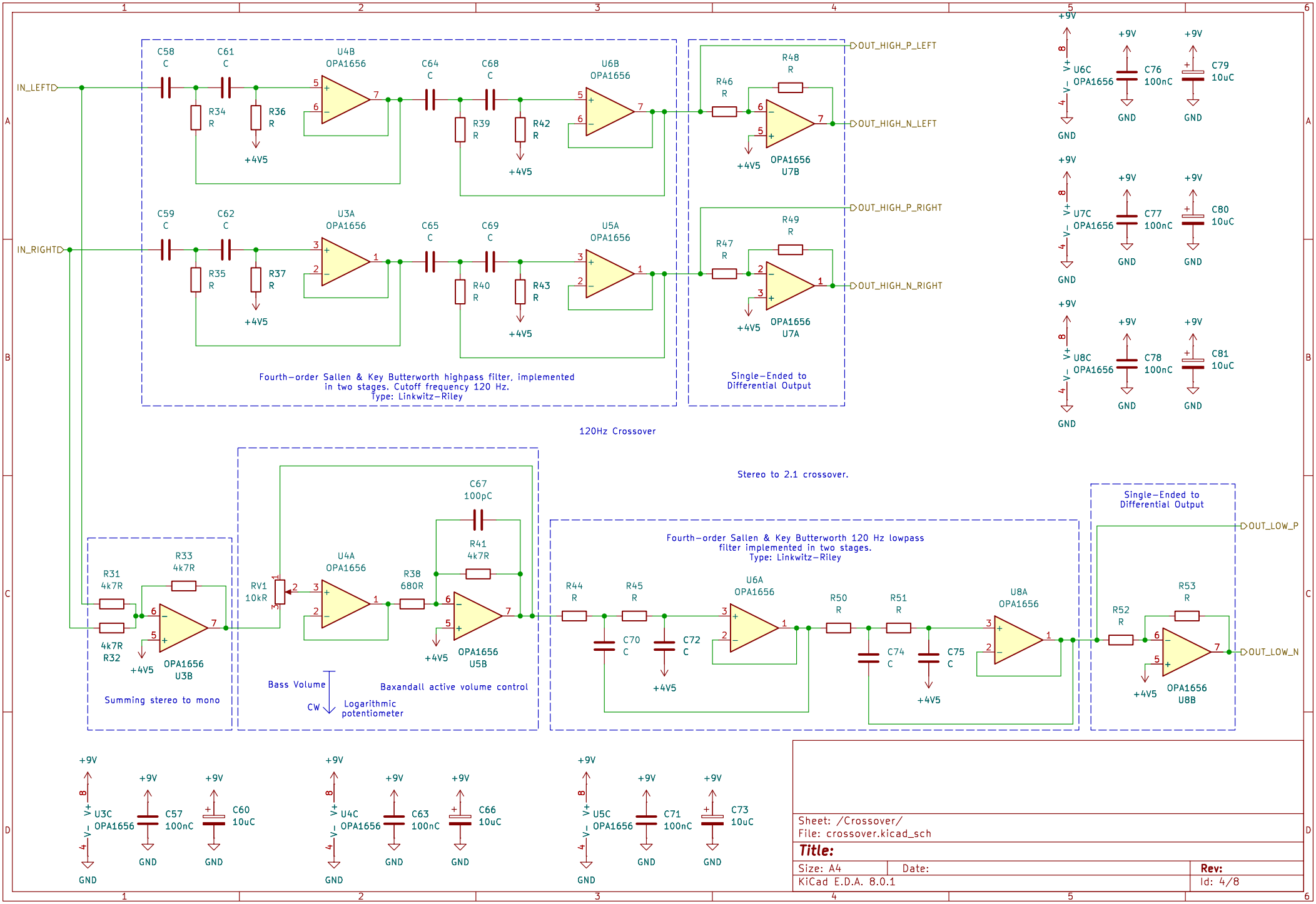


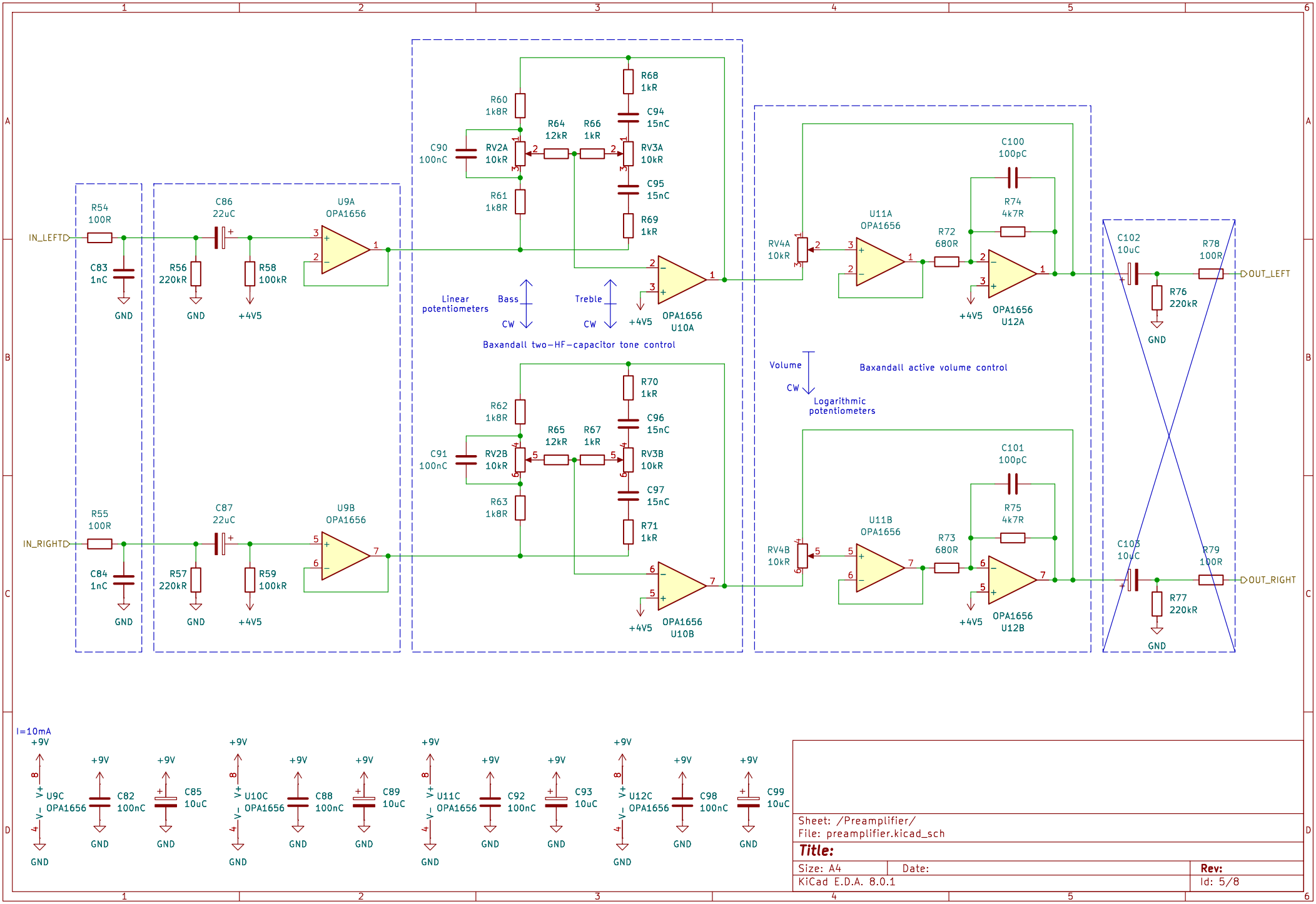


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Size: A4	Date:	**Rev:**
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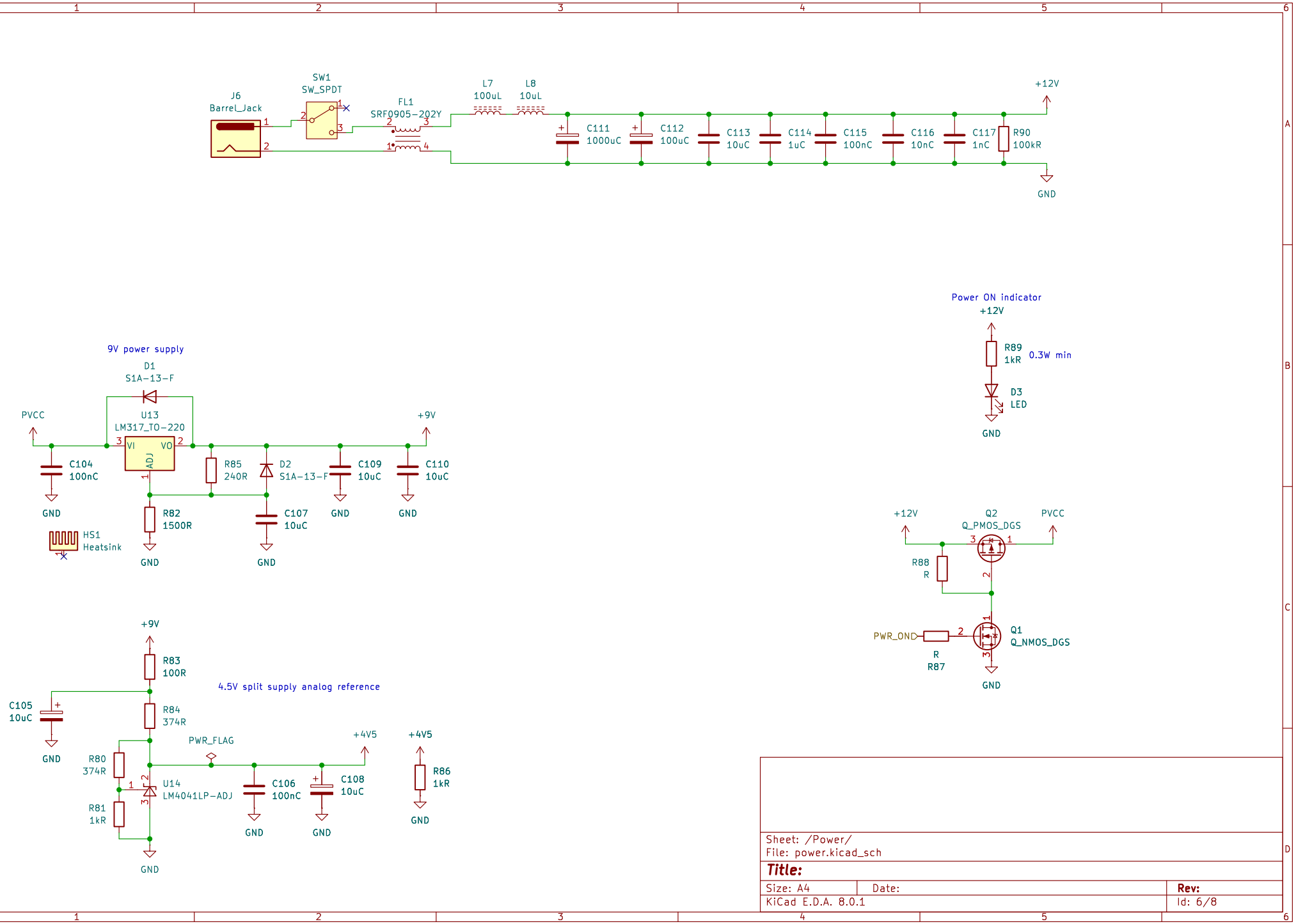


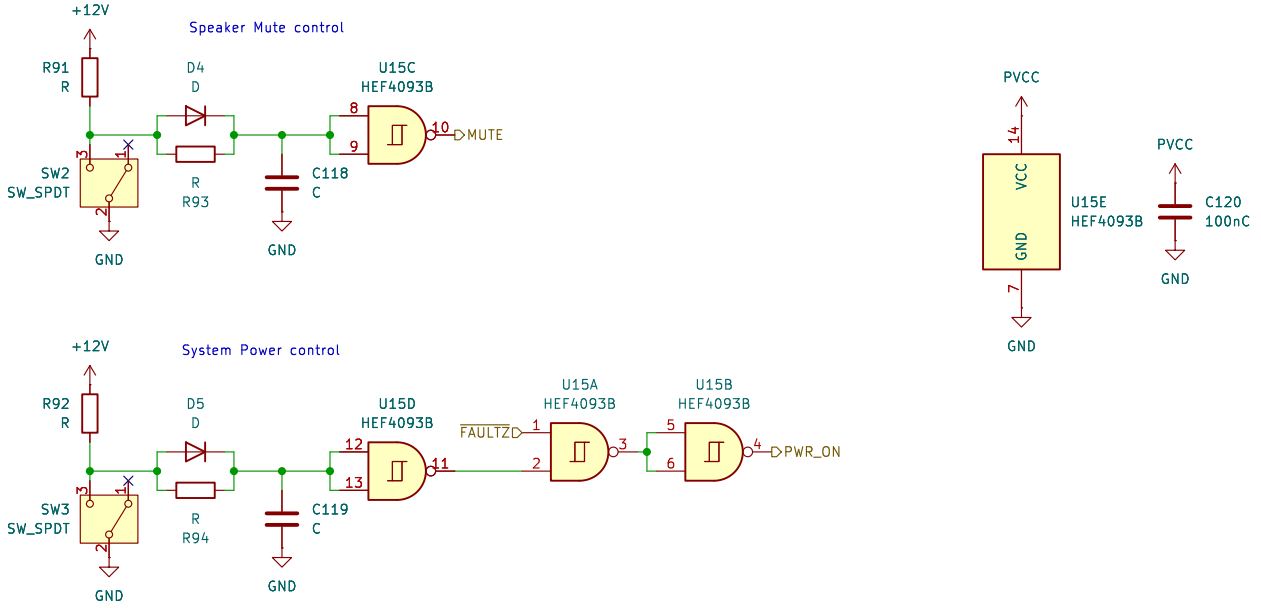
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File: mono_amp.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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IN_LEFT
IN_RIGHT
C58 C
C61 C
C59 C
C62 C
U4B OPA1656
U3A OPA1656
C64 C
C68 C
C65 C
C69 C
U6B OPA1656
U5A OPA1656
R34 R
R36 R
R35 R
R37 R
R39 R
R42 R
R40 R
R43 R
+4V5
Fourth-order Sallen & Key Butterworth highpass filter, implemented in two stages. Cutoff frequency 120 Hz.
Type: Linkwitz-Riley
R46 R
R48 R
R47 R
R49 R
OPA1656 U7B
OPA1656 U7A
OUT_HIGH_P_LEFT
OUT_HIGH_N_LEFT
OUT_HIGH_P_RIGHT
OUT_HIGH_N_RIGHT
Single-Ended to Differential Output
+9V
GND
U6C OPA1656
U7C OPA1656
U8C OPA1656
C76 100nC
C77 100nC
C78 100nC
C79 10uC
C80 10uC
C81 10uC
120Hz Crossover
Stereo to 2.1 crossover.
R31 4k7R
R33 4k7R
4k7R R32
OPA1656 U3B
Summing stereo to mono
RV1 10kR
U4A OPA1656
R38 680R
C67 100pC
R41 4k7R
OPA1656 U5B
Bass Volume
CW
Logarithmic potentiometer
Baxandall active volume control
Fourth-order Sallen & Key Butterworth 120 Hz lowpass filter implemented in two stages.
Type: Linkwitz-Riley
R44 R
R45 R
U6A OPA1656
R50 R
R51 R
U8A OPA1656
C70 C
C72 C
C74 C
C75 C
Single-Ended to Differential Output
R52 R
R53 R
OPA1656 U8B
OUT_LOW_P
OUT_LOW_N
U3C OPA1656
U4C OPA1656
U5C OPA1656
C57 100nC
C63 100nC
C71 100nC
C60 10uC
C66 10uC
C73 10uC
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File: crossover.kicad_sch
Title:
Size: A4
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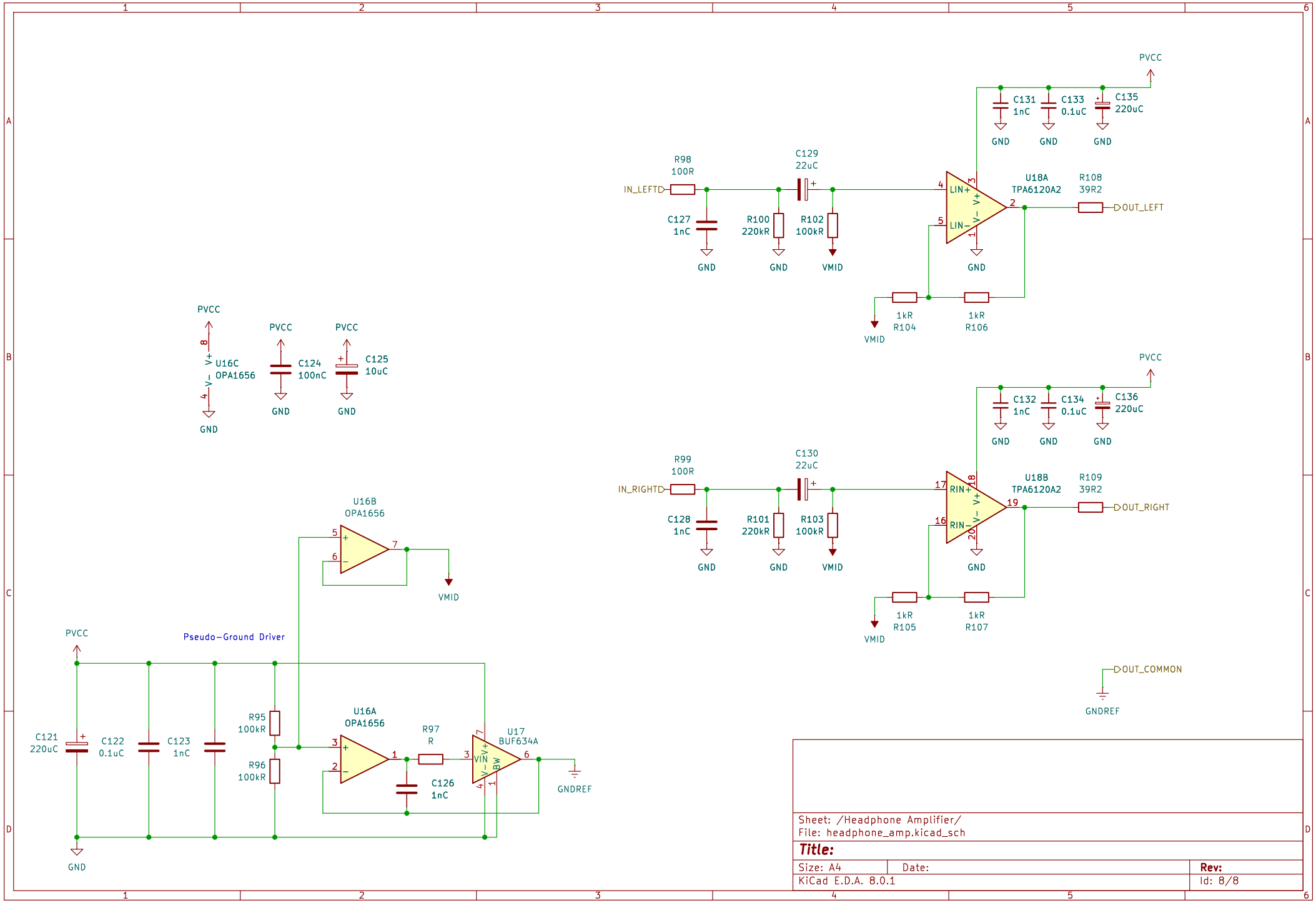


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File: preamplifier.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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J6
Barrel_Jack
SW1
SW_SPDT
FL1
SRF0905-202Y
L7
100uL
L8
10uL
+12V
C111
1000uC
C112
100uC
C113
10uC
C114
1uC
C115
100nC
C116
10nC
C117
1nC
R90
100kR
GND
9V power supply
D1
S1A-13-F
PVCC
U13
LM317_TO-220
VI
VO
ADJ
+9V
C104
100nC
R85
240R
D2
S1A-13-F
C109
10uC
C110
10uC
R82
1500R
C107
10uC
HS1
Heatsink
Power ON indicator
R89
1kR
0.3W min
D3
LED
+12V
Q2
Q_PMOS_DGS
PVCC
R88
R
PWR_ON
Q1
Q_NMOS_DGS
R87
R83
100R
4.5V split supply analog reference
C105
10uC
R84
374R
PWR_FLAG
+4V5
R80
374R
U14
LM4041LP-ADJ
C106
100nC
C108
10uC
R86
1kR
R81
1kR
Sheet: /Power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
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Sheet: /Control/		
File: control.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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PVCC
C131
1nC
C133
0.1uC
C135
220uC
GND
R98
100R
IN_LEFT
C129
22uC
C127
1nC
R100
220kR
R102
100kR
VMID
U18A
TPA6120A2
R108
39R2
OUT_LEFT
LIN+
LIN−
V+
V−
1kR
R104
1kR
R106
C132
1nC
C134
0.1uC
C136
220uC
R99
100R
IN_RIGHT
C130
22uC
C128
1nC
R101
220kR
R103
100kR
U18B
TPA6120A2
R109
39R2
OUT_RIGHT
RIN+
RIN−
1kR
R105
1kR
R107
OUT_COMMON
GNDREF
U16C
OPA1656
C124
100nC
C125
10uC
U16B
OPA1656
Pseudo−Ground Driver
C121
220uC
C122
0.1uC
C123
1nC
R95
100kR
R96
100kR
U16A
OPA1656
R97
R
C126
1nC
U17
BUF634A
VIN
BW
Sheet: /Headphone Amplifier/
File: headphone_amp.kicad_sch
Title:
Size: A4
Date:
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KiCad E.D.A. 8.0.1
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